Key Points

Course: Social Network Analysis, UIT, VNUHCM
(aka. Graph and Social Media Analysis with Applications)

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Topic 3: Measuring Networks & Network Structure

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1. Degree and Degree Sequence

The **degree** of a node measures how many connections it has.

- Undirected graph:

$$k_v = |\Gamma(v)|$$

- Directed graph:
  - In-degree: k_v^{in}
  - Out-degree: k_v^{out}

The **degree sequence** is the list $(k_1, k_2, \dots, k_n)$.

Example

Graph with edges: (A,B), (A,C), (A,D)

- $k_A = 3$
- $k_B = k_C = k_D = 1$

Notes

- Degree is the most local network property.
- Many later concepts (centrality, link prediction, diffusion) depend directly on degree.

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2. Handshaking Lemma and Average Degree

For an undirected graph with m edges:

$$\sum_{v \in V} k_v = 2m$$

The **average degree** is:

$$\langle k \rangle = \frac{1}{n} \sum_{v} k_v = \frac{2m}{n}$$

For directed graphs:

- $\sum_v k_v^{in} = \sum_v k_v^{out} = m$

Intuition

Each edge contributes degree 1 to each of its endpoints (undirected).

Notes

- Useful for quick sanity checks.
- Average degree does *not* describe heterogeneity.

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3. Degree Distribution

The **degree distribution** $P(k)$ is defined as:

$$P(k) = \frac{\#\{v : k_v = k\}}{n}$$

It tells us the probability that a randomly chosen node has degree k.

Intuitively, the degree distribution shows how many nodes have degree 0, 1, 2, 3, ...

Example

Degrees: $(1, 1, 1, 2, 3)$ Then:

- $P(1) = 3/5$
- $P(2) = 1/5$

- $P(3) = 1/5$

Notes

- $P(k)$ is usually visualized as a histogram or log-log plot.
- Degree distribution is a **global** structural property.

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4. Heavy-Tailed and Power-Law Degree Distributions

Many real networks exhibit **heavy-tailed** degree distributions:

- Vast majority of the nodes have low degree.
- A few nodes (hubs) have very high degree.

Specifically, they usually follow the scale-invariant **power-law** distribution, with the form:

$$P(k) \propto k^{-\gamma}$$

for some exponent $\gamma > 1$.

Intuition

There is no “typical” degree scale; hubs dominate connectivity.

Examples

- Social networks
- Web graph
- Citation networks

Notes

- Not all heavy-tailed distributions are true power laws.
- Log-log plots help reveal approximate power-law behavior.
- Fitting power laws requires care; naive fitting is often wrong.

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5. Log-Log Plots of Degree Distributions

A **log-log plot** plots:

$$(\log k, \log P(k))$$

If $P(k)$ follows a power law, the plot appears approximately linear with slope $-\gamma$.

Intuition

Log-log plots compress wide ranges of values and highlight tail behavior.

Notes

- Straight line ≠ proof of power law.
- Finite-size effects and noise are common in real data.

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6. Paths, Distances, and Shortest Paths

The **distance** $d(u, v)$ is the length of the shortest path between nodes u and v.

- If no path exists, distance is infinite.
- In weighted graphs, distance is sum of edge weights.

The **average shortest path length** is:

$$\ell = \frac{1}{n(n-1)} \sum_{u \neq v} d(u, v)$$

(usually computed over connected pairs).

Example

Chain A-B-C-D:

- $d(A, D) = 3$

Notes

- Distance underlies closeness centrality and diffusion speed.
- Real networks often have surprisingly small average path lengths.

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7. Small-World Property

A network exhibits the **small-world property** if it satisfies both:

1. **Short average path length**, comparable to a random graph.
2. **High clustering coefficient**, much higher than a random graph.

Intuition

Networks are locally clustered but globally well-connected.

Example

Social networks:

- Your friends know each other (high clustering).
- You can reach distant people in a few hops.

Notes

- Small-world ≠ scale-free.
- Watts–Strogatz model demonstrates this phenomenon.

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8. Local Clustering Coefficient

The **local clustering coefficient** of node v is:

$$C_v = \frac{2T_v}{k_v(k_v - 1)}$$

where:

- T_v = number of triangles containing v
- k_v = degree of v

Defined only when $k_v \geq 2$.

Example

If node v has 3 neighbors and 2 edges among them:

- $C_v = 2 \cdot 2/(3 \cdot 2) = 2/3$

Intuition

Measures how close the neighbors of a node are to forming a clique.

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9. Global Clustering Coefficient

Two common definitions:

1. **Average local clustering**:

$$C = \frac{1}{n} \sum_v C_v$$

2. **Transitivity**:

$$C = \frac{3 \times (\text{number of triangles})}{\text{number of connected triples}}$$

Notes

- Both capture “triadic closure”.
- They can differ numerically, especially in heterogeneous graphs.

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10. Random Graph Models (Erdős–Rényi)

The **Erdős–Rényi (ER) model** $G(n, p)$:

- n nodes.
- Each possible edge appears independently with probability p.

Expected properties:

- Expected degree: $\langle k \rangle = p(n - 1)$
- Degree distribution: approximately binomial (or Poisson for large n, small p).

Intuition

Acts as a baseline for comparison with real networks.

Notes

- ER graphs have low clustering.
- ER graphs do not produce hubs naturally.

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11. Comparing Real Networks with Random Graphs

There are 4 key network properties:

- Degree distribution
- Average path length
- Clustering coefficient
- Connected components & Community structure

Real networks usually follow Small-World model, differring from ER random graphs on some key properties:

Property	Real/Small-World	ER Random Graph	Matches real networks?
Degree distribution	Heavy-tailed	Bell-shaped	✗
Average path length	Small	Small	✓
Clustering coefficient	High	Low	✗
Community structure	Strong	Weak (only connected if $\langle k \rangle > 1$)	✗ (✓ if $\langle k \rangle > 1$)

Intuition

Random graphs explain connectivity but not structure.

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12. Sampling Bias and Measurement Pitfalls

Observed degree distributions may be distorted by:

- BFS or snowball sampling
- Missing nodes or edges
- API rate limits

Notes

- Sampling bias can artificially create heavy tails.
- Always question how data was collected.

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13. Why Network Structure Matters

Structural properties influence:

- Speed of information diffusion
- Robustness to node/edge removal
- Effectiveness of centrality measures
- Accuracy of link prediction

Intuition

Structure determines dynamics. This motivates later lectures on diffusion and learning.

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14. Summary of Structural Metrics Introduced

This topic introduces metrics at different scales:

- **Local**: degree, local clustering
- **Node-pair**: shortest paths
- **Global**: degree distribution, average path length, global clustering
- **Model-based**: comparison with random graphs

These form the foundation for centrality, communities, and diffusion.
